

Vishal Singh

COMPUTER SCIENCE ENGINEER

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Live Portfolio (Click Here)

SKILLS MATRIX

LANGUAGES

Python | Java | C++ | JavaScript | SQL

AI & MACHINE LEARNING

Scikit-learn | OpenCV | Pandas | NumPy | Data Modeling

FULL STACK & BACKEND

HTML5 / Modern CSS3 | Vanilla ES6+ | Flask | FastAPI | REST APIs

SECURITY & SYSTEMS

Zero-Trust Arch | eBPF XDP | Shannon Entropy | Linux CLI | Git / GitHub

EDUCATION

B.Tech in Computer Science

VISHAL SINGH RAJPUROHIT

AVAILABLE IMMEDIATELY

Computer Science Engineer | AI & Machine Learning | Full-Stack Systems

EXECUTIVE SUMMARY

High-performing **Computer Science Engineer (CGPA 8.9/10)** with deep technical rigor across Artificial Intelligence, Machine Learning pipelines, and sub-millisecond full-stack architectures. Proven track record of independently designing, building, and deploying **5+ production-grade software applications** ranging from computer vision smart systems to AI-powered cybersecurity anomaly detectors. Known for strong algorithmic problem solving, clean code architecture, and zero-framework optimization.

FEATURED PRODUCTION PROJECTS

1. AI Cyber Threat & Packet Anomaly Detector (NIDS)

`Python · Scikit-learn · eBPF · Flask · Security`

- Engineered a real-time Network Intrusion Detection System (NIDS) analyzing high-velocity packet streams.
- Implemented **Shannon Information Entropy calculation ($H > 7.5$)** to automatically flag zero-day encrypted payloads.
- Designed automated defensive response specifications utilizing **eBPF XDP filter drivers** to mitigate SYN flood anomalies at the network driver ring with zero userspace overhead.

2. AI Electricity Consumption Optimizer

`Machine Learning · Pandas · Time-Series · Python`

- Architected an intelligent predictive model analyzing time-series energy telemetry to forecast consumption patterns.
- Trained anomaly detection algorithms identifying uncharacteristic spikes and energy leakage across dynamic load conditions.
- Delivered automated energy conservation recommendations resulting in a projected **18% reduction** in power overhead.

3. FinTech Insurance Fraud Detection System

`Random Forest · Data Modeling · Flask API · SQL`

- Built an end-to-end fraud classifier predicting fraudulent insurance claims with a **94.2% test accuracy rate**.

Bapatla Engineering College

2021 – 2025 | Full-Time

★ CGPA: 8.9 / 10.0

VERIFIED BADGES

- ✓ HackerRank Python (Gold Badge)
- ✓ Problem Solving Certified (5★)
- ✓ 150+ LeetCode DSA Solutions
- ✓ NPTEL Certified in Core Algorithms

- Processed multi-dimensional customer behavioral vectors, policy claim histories, and transactional anomaly features.
- Exposed high-throughput REST API endpoints consumed by an interactive analytical visualization dashboard.

4. Smart Attendance System via Computer Vision

`OpenCV · Deep Facial Embeddings · SQLite · Python`

- Developed an automated facial recognition attendance management pipeline processing video feeds at **60 FPS**.
- Implemented face detection, landmark alignment, and biometric feature comparison against relational student records.
- Automated daily SQL record generation with anti-spoofing checks and zero manual administrative intervention.

5. High-Performance URL Shortener & Analytics

`Vanilla JS · Python Flask · Relational DB · QR Engine`

- Built a full-stack link shortening platform delivering **sub-5 millisecond redirection latency**.
- Equipped with real-time geo-telemetry tracking, device/browser analytics, and dynamic vector QR code generators.

COMPETITIVE RIGOR & MILESTONES

150+ DSA Problems Solved
Demonstrated algorithmic mastery across LeetCode, HackerRank, and competitive programming platforms.

Zero-Framework Architecture
Engineered an interactive developer portfolio scoring 100/100 Lighthouse performance with pure HTML/CSS/JS.

Academic Distinction (8.9 CGPA)
Consistent top-tier academic standing across 8 semesters at Bapatla Engineering College.

AI Sentinel Assistant Builder
Architected custom NLP conversational co-pilot deployed live on GitHub Pages for recruiter queries.